

Fork-Merge Decoding: Enhancing Multimodal Understanding in Audio-Visual Large Language Models

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BUDAPEST

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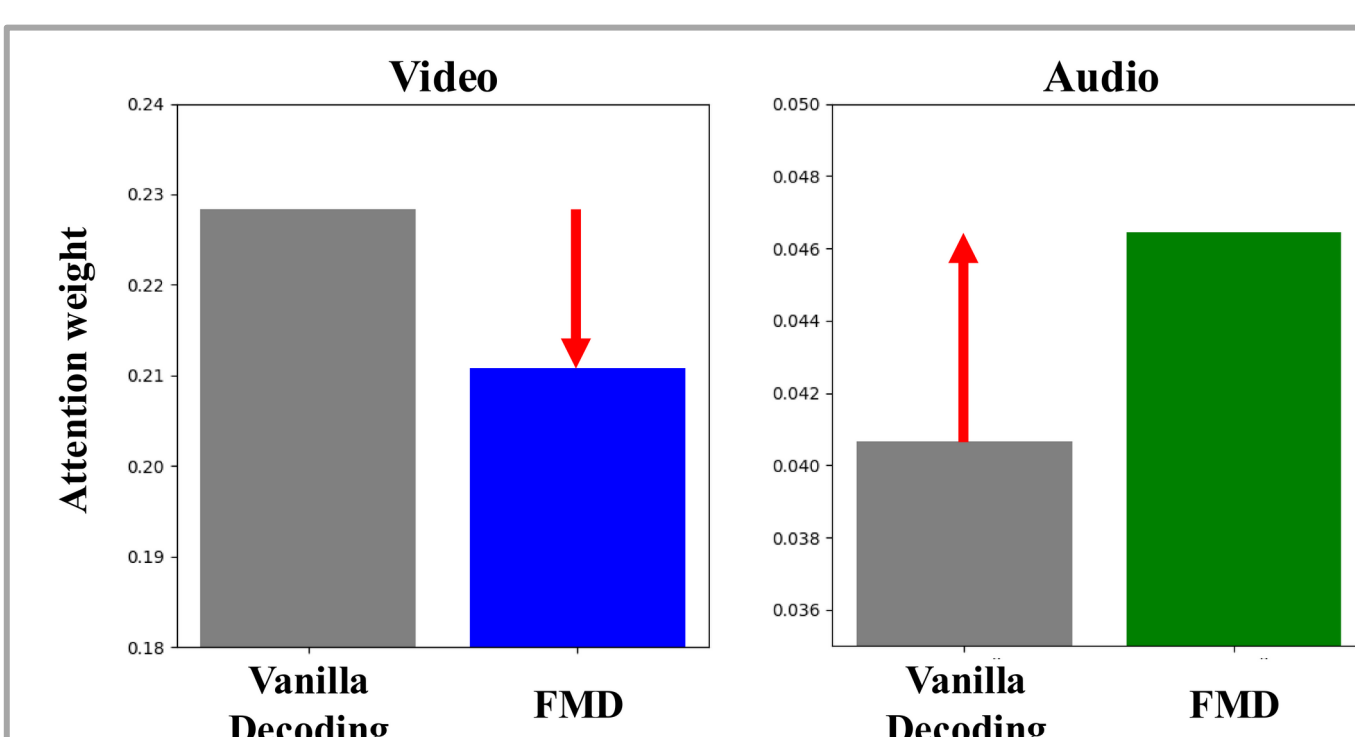
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Motivation

AV-LLMs can exhibit a clear **modality bias** toward **video**.

We hypothesize that this stems from **data bias**: video-text pairs are abundant, whereas audio-video-text pairs are scarce.



Where does the final question token attend?
Last decoder layer · VideoLLaMA2 · 100 AVHBench samples

- Vanilla: attention concentrated on visual features over audio.
- FMD effectively mitigates this bias.

Key Idea

Separate first, integrate later.

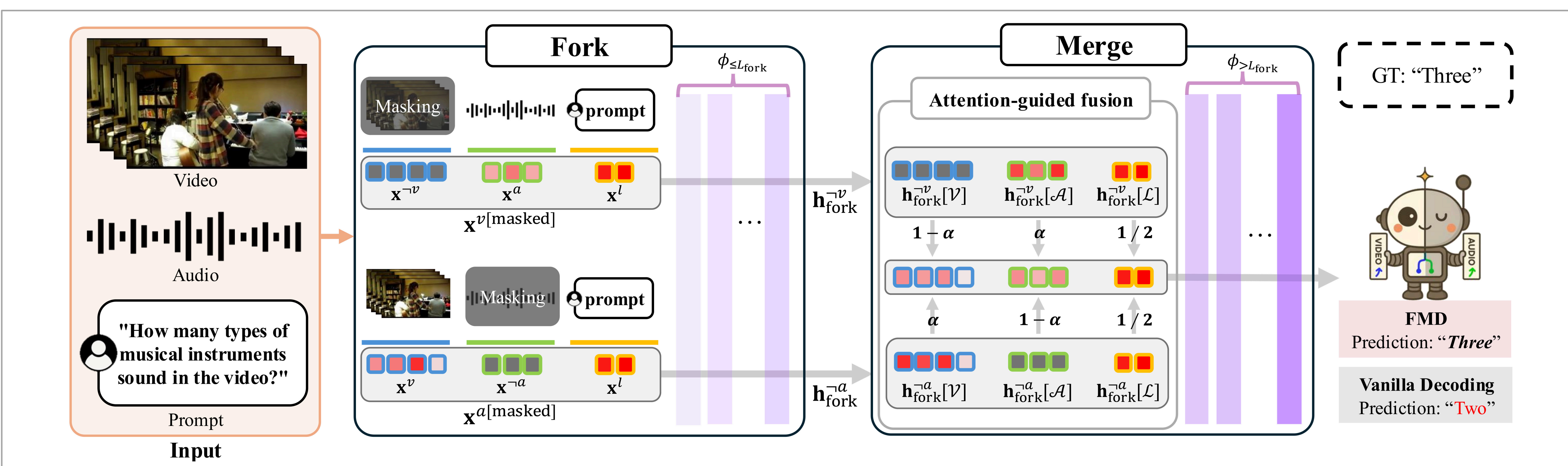
Fork modalities in early layers, **merge** them for joint reasoning!

- Fork**: mask video or audio (keep the question) for **modality-specific reasoning** in early layers.
- Merge**: combine both branches for **cross-modal interaction and reasoning** in later layers.

Training-free

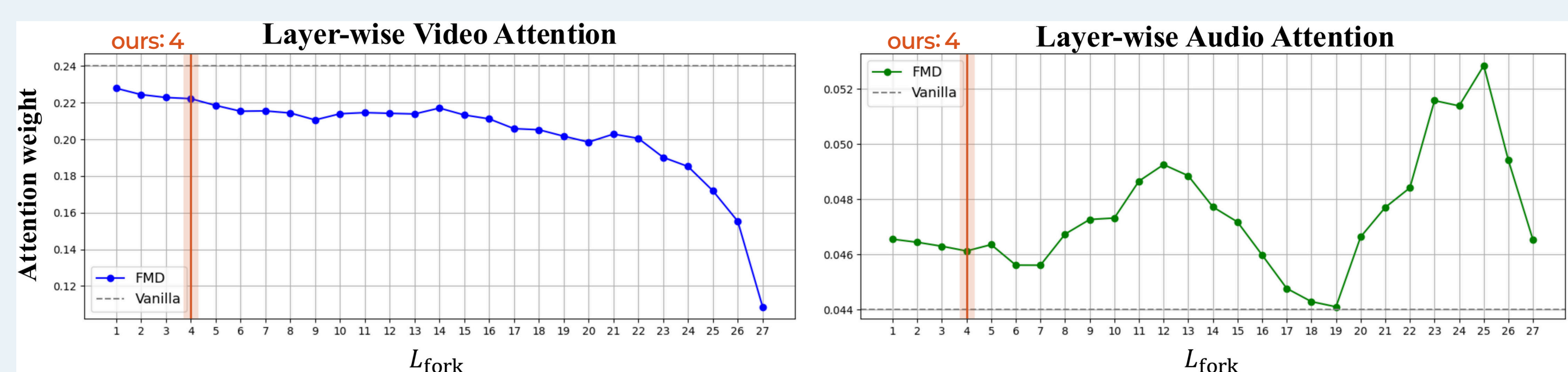
Plug-and-play

Fork-Merge Decoding (FMD)



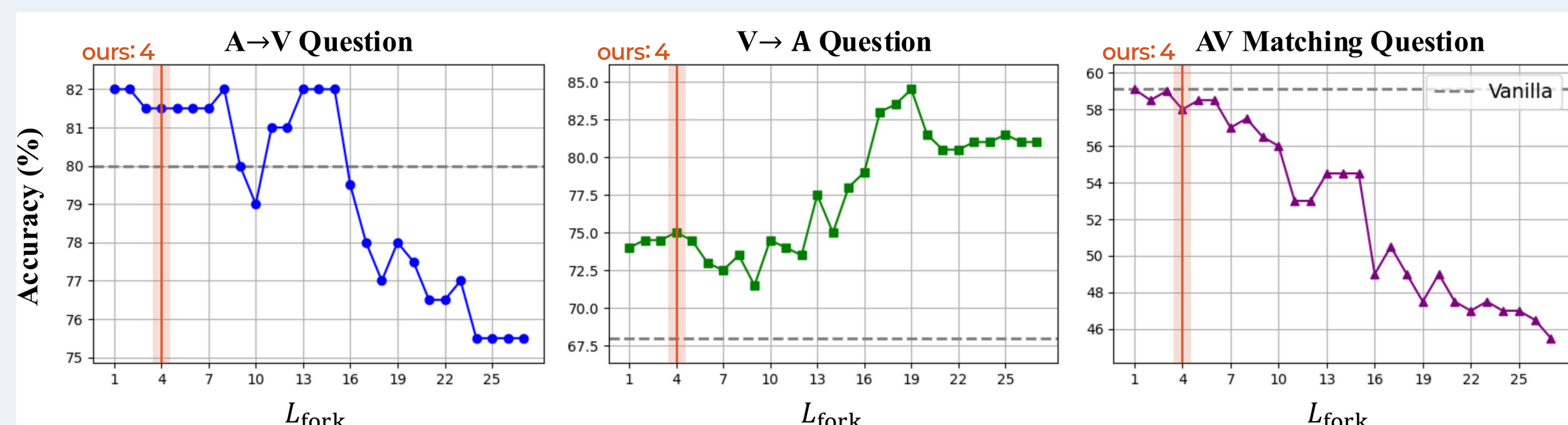
Analysis: FMD Reduces Attention Imbalance

Increasing L_{fork} shifts attention from video to audio (VideoLLaMA2)



- An early fork already lowers video attention and raises audio attention.

Too deep hurts vision: fork early



- Deep forks improve V→A but hurt A→V and AV matching.

Efficiency: Only Early Layers Run Twice

vs. test-time decoding methods (VideoLLaMA2, AVHBench)

Decoding	AVHBench accuracy (%) ↑			Relative latency ↓
	A→V	V→A	AV Mat.	
Vanilla	80.02	77.03	57.75	1.00×
DoLa (Chuang et al., 2024)	69.34	63.44	48.33	1.59×
VCD (Leng et al., 2024)	75.96	69.67	52.52	2.03×
SID (Huo et al., 2025)	78.53	72.82	53.52	2.09×
FMD (Ours)	80.45	77.52	59.01	1.26×

- Higher accuracy than SID / VCD ($\approx 2\times$) at only **1.26×** latency.

Main Results

Benchmark	Model	VideoLLaMA2		video-SALMONN		Qwen2.5-Omni	
		Vanilla	FMD	Vanilla	FMD	Vanilla	FMD
AVHBench							
A→V		80.02	80.45	68.69	70.51	80.77	81.30
V→A		77.03	77.52	62.39	65.41	71.20	71.77
AV matching		57.75	59.01 +1.26	49.46	54.77 +5.31	77.45	78.22 +0.77
AV captioning		2.84	2.95	1.83	2.01	3.25	3.36
Audio-Visual QA & Video Understanding							
AVQA		60.23	61.46	28.20	40.46	86.49	86.61
MUSIC-AVQA		81.30	81.50	44.48	50.60	N/A	N/A
AVUT		44.44	45.13	32.36	32.85	N/A	N/A
WorldSense		29.41	29.91	56.20	57.83	N/A	N/A

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All 21 comparisons in the table improve over vanilla

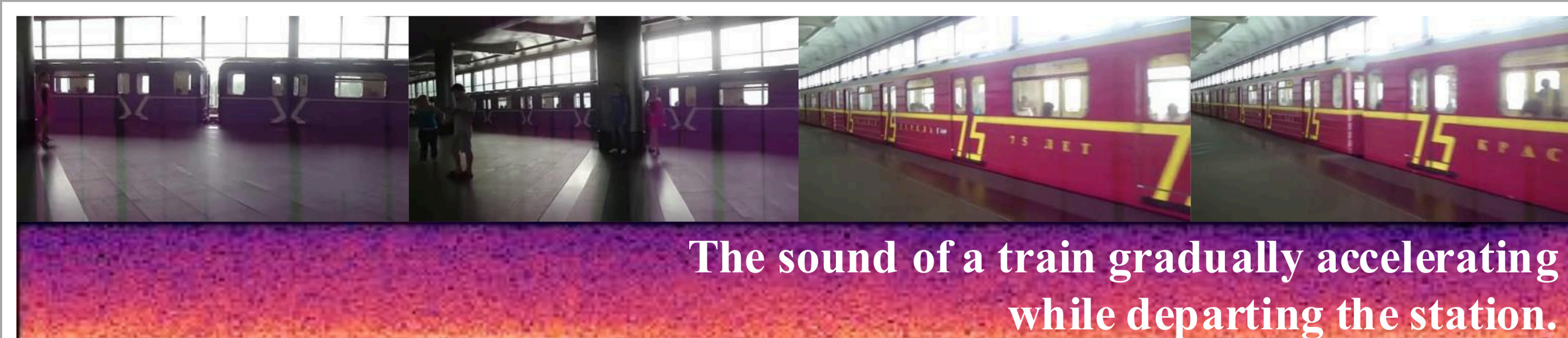
+5.31 pp

AV matching, video-SALMONN
49.46 → 54.77

+12.26 pp

AVQA, video-SALMONN (zero-shot)
28.20 → 40.46

Qualitative Result (VideoLLaMA2, AVQA)



Q. What's driving in the video? (A) Cable car (B) Eat fruit (C) Train (D) High-speed rail

Vanilla: (A) Cable car

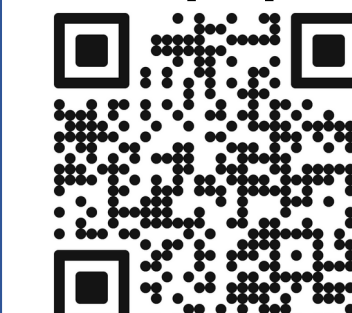
FMD: (C) Train (correct)

Audio resolves the visual ambiguity.

Conclusion

- FMD**: separate early, merge later, **without training**.
- Mitigates modality bias; improves **all main results**.
- Only early layers run twice: **1.26×** latency.

Our paper



Q&A notes

